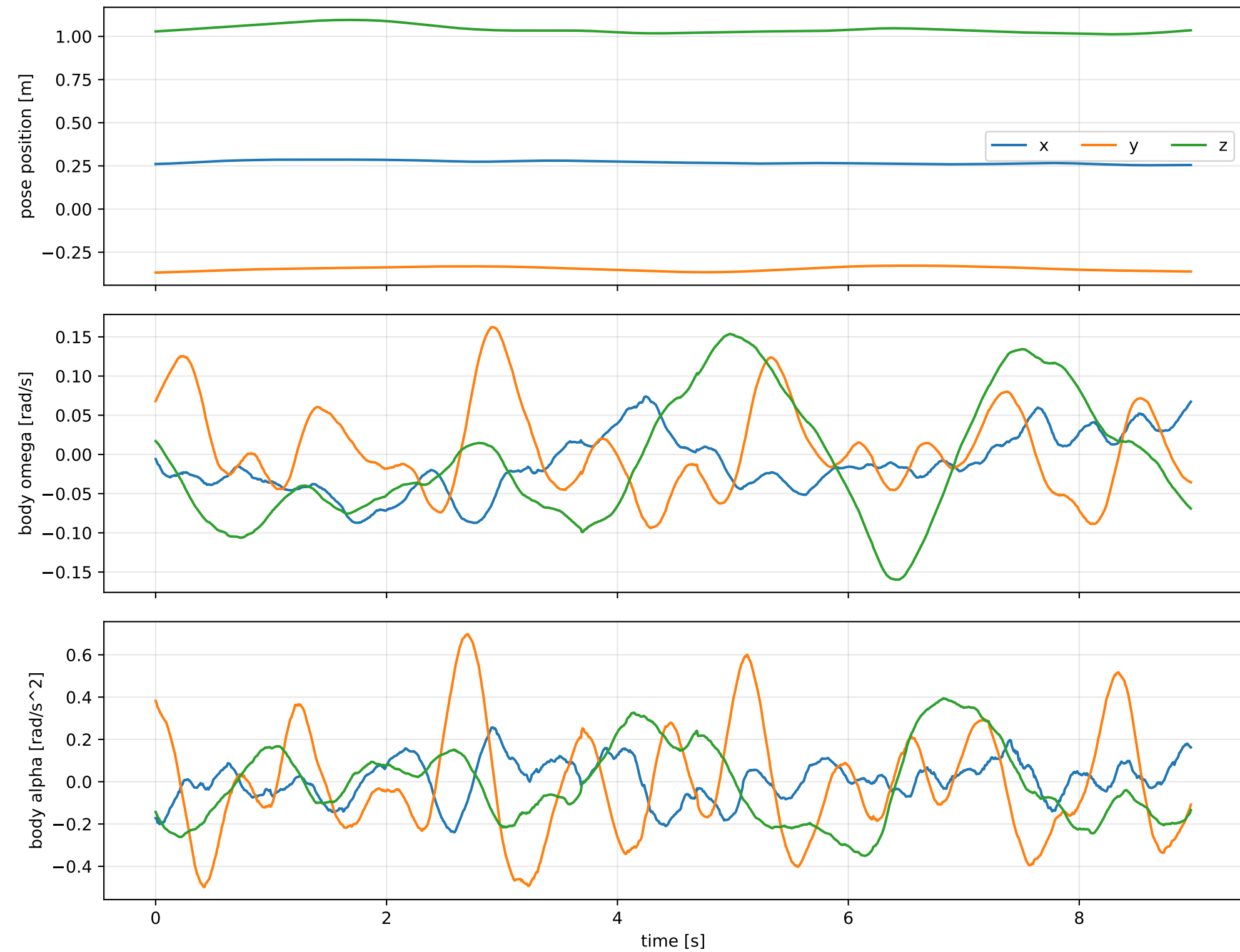
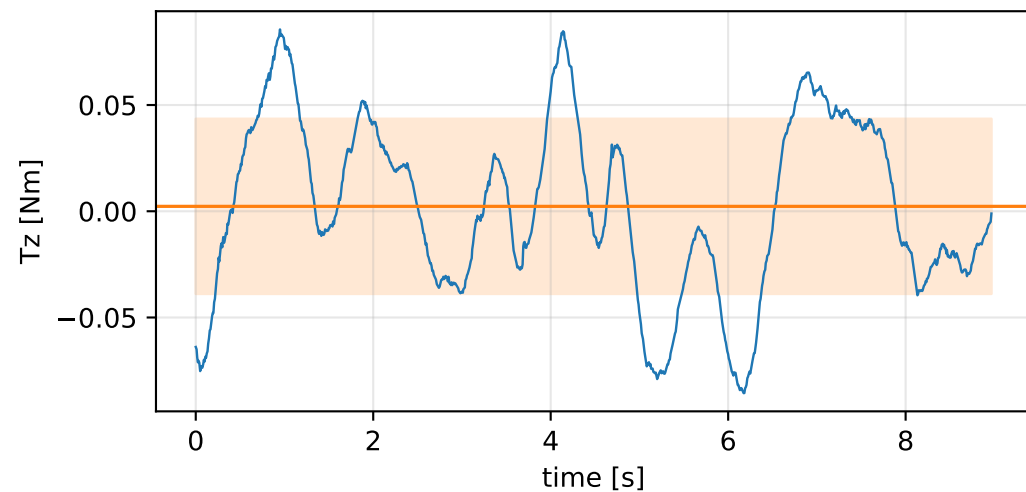
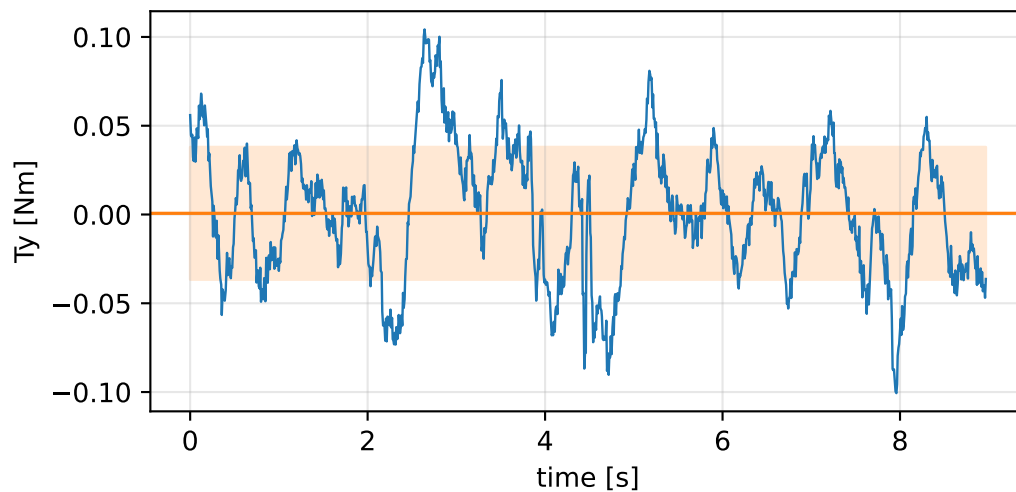
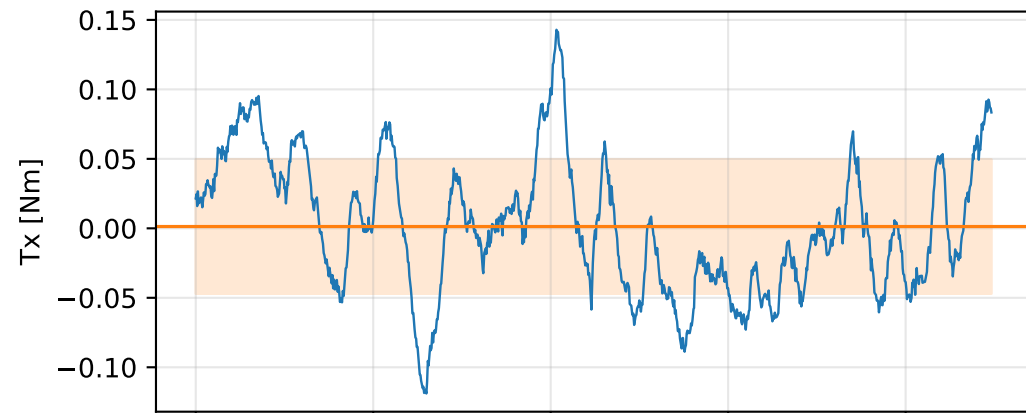
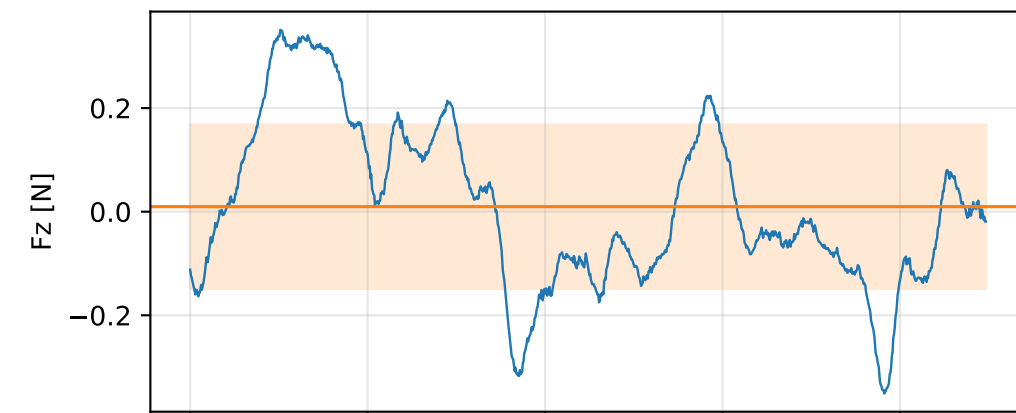
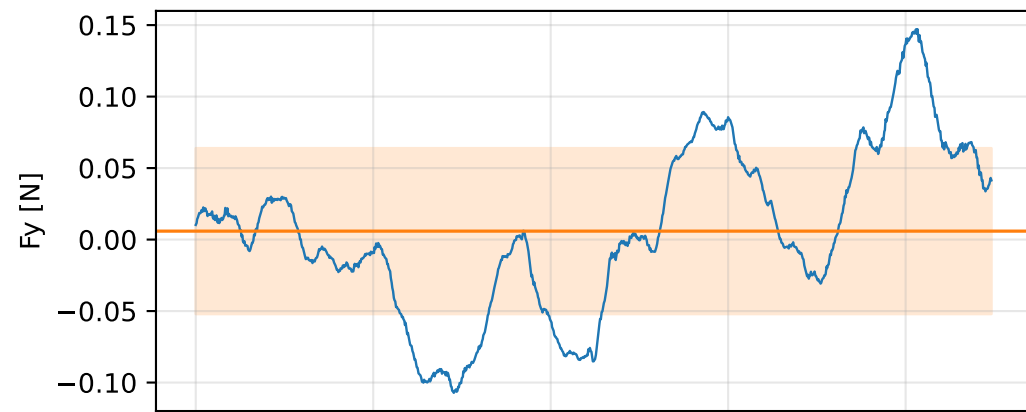
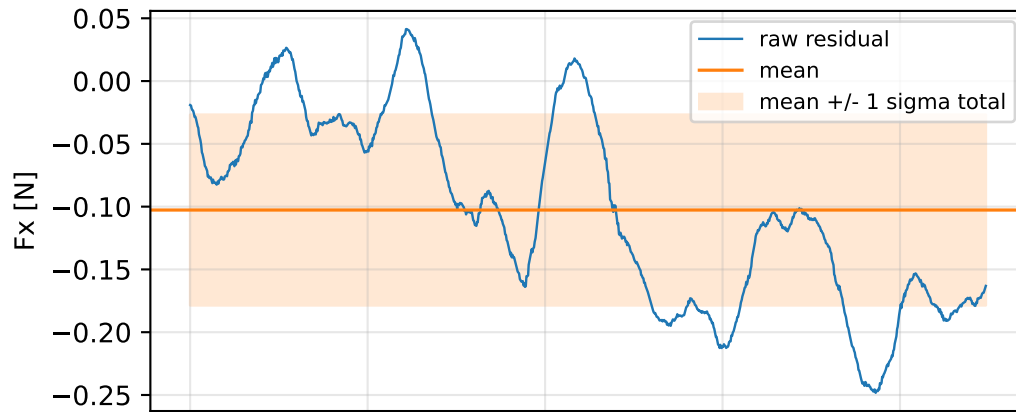


force_over_mass_rotor_2_nominal: SG trajectory and derived motion



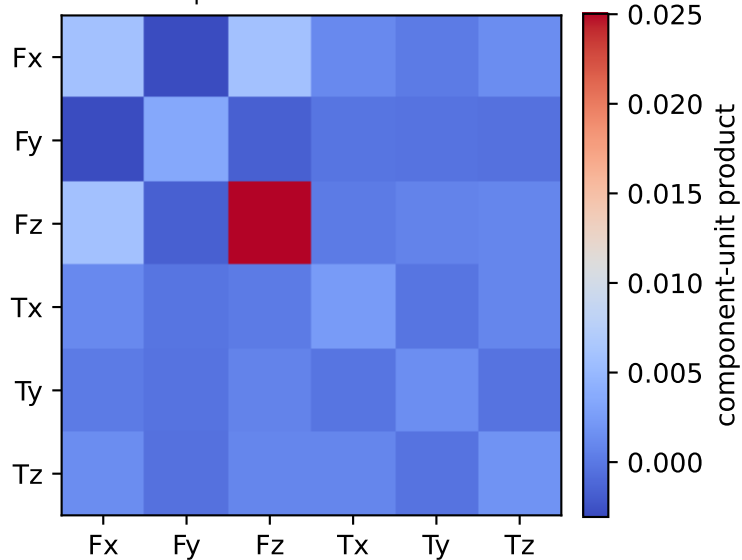
force_over_mass_rotor_2_nominal: raw Newton--Euler residual wrench (not trajectory-fitted external wrench)
mass gauge fixed to nominal mass = 2.35159759 kg



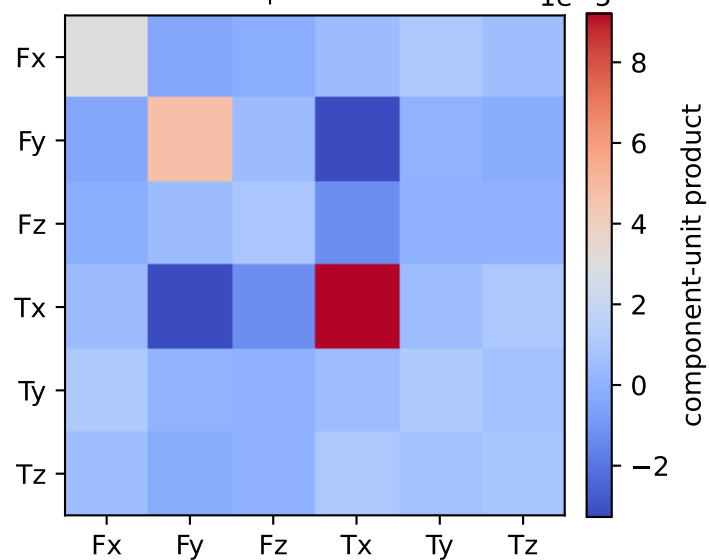
vector RMS about zero: force 0.21163 N, torque 0.073637 Nm; component std about mean: force [0.0759 0.058 0.1582], torque [0.0483 0.0373 0.0412]

force_over_mass_rotor_2_nominal: residual-wrench covariance decomposition in nominal-mass gauge
force [N], torque [Nm]; raw excess eigenvalues [0.001 0.001 0.001 0.003 0.007 0.027]

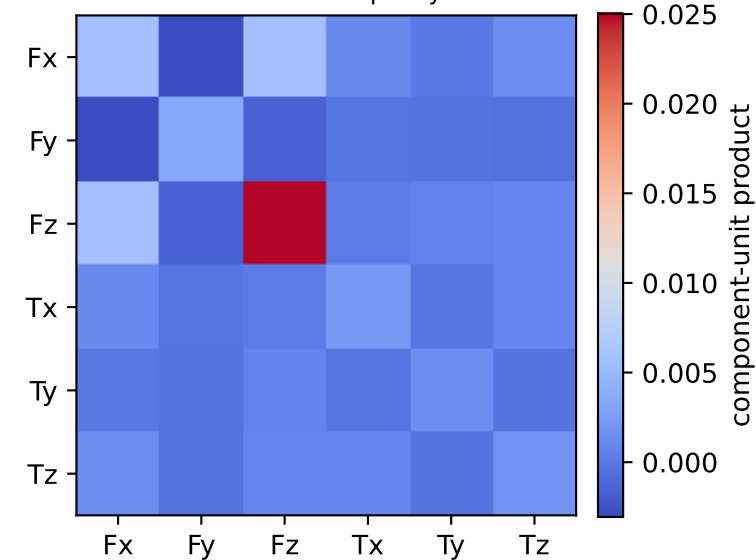
empirical total covariance



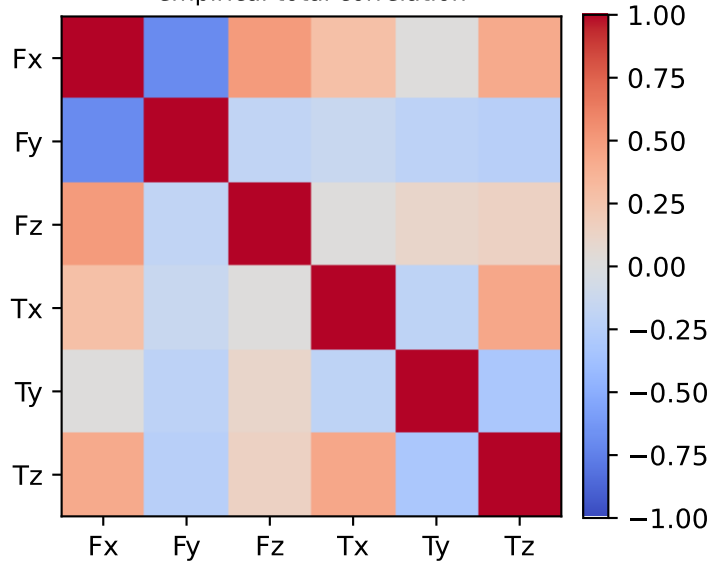
mean full-SG prediction covariance $1e-5$



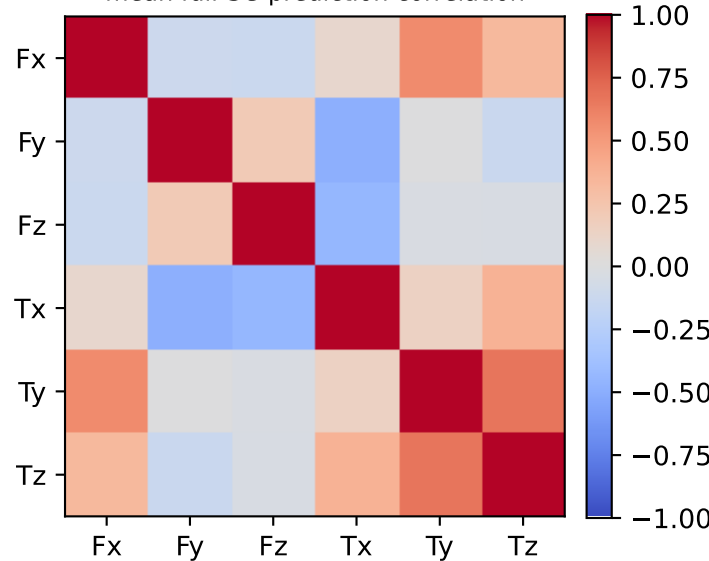
PSD excess model discrepancy covariance



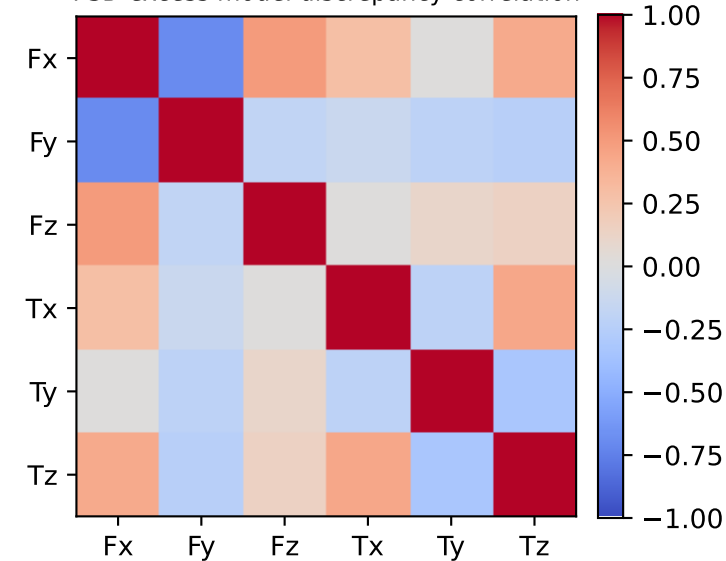
empirical total correlation



mean full-SG prediction correlation

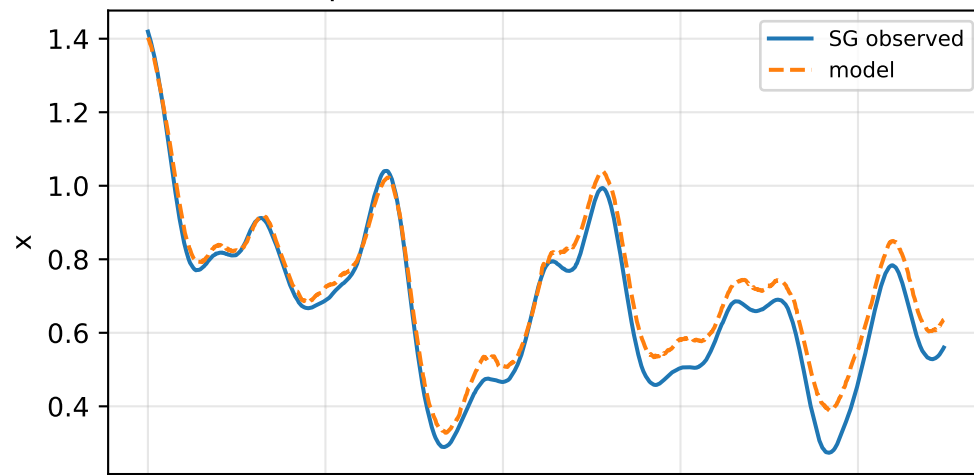


PSD excess model discrepancy correlation

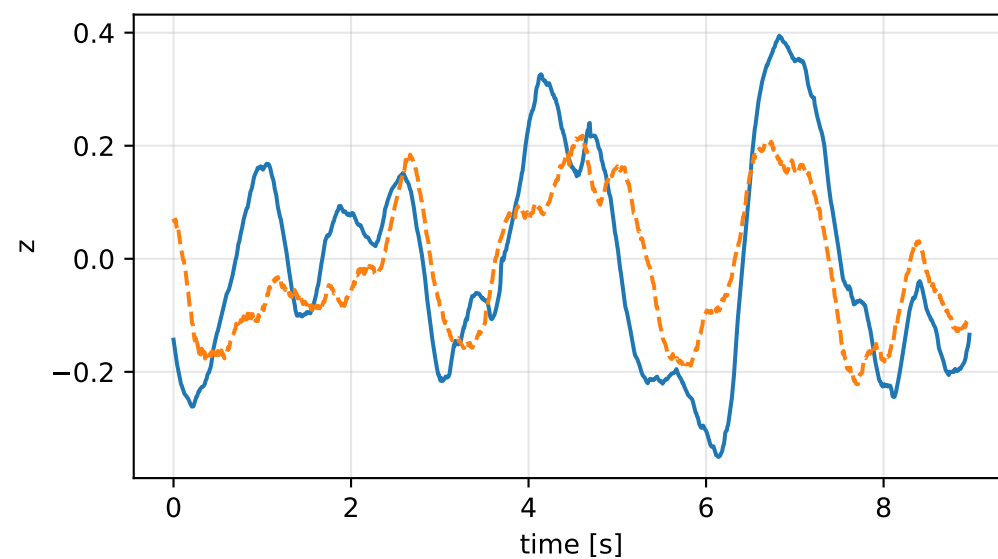
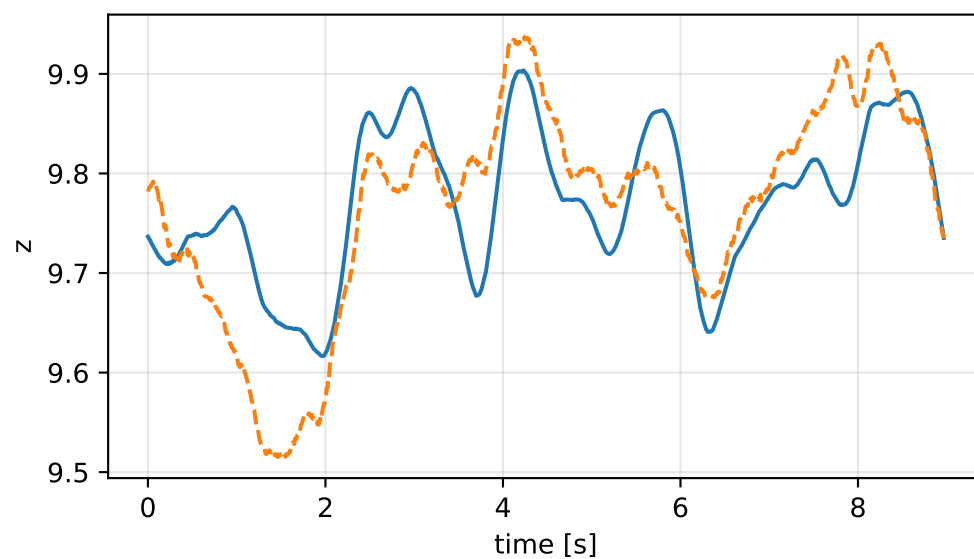
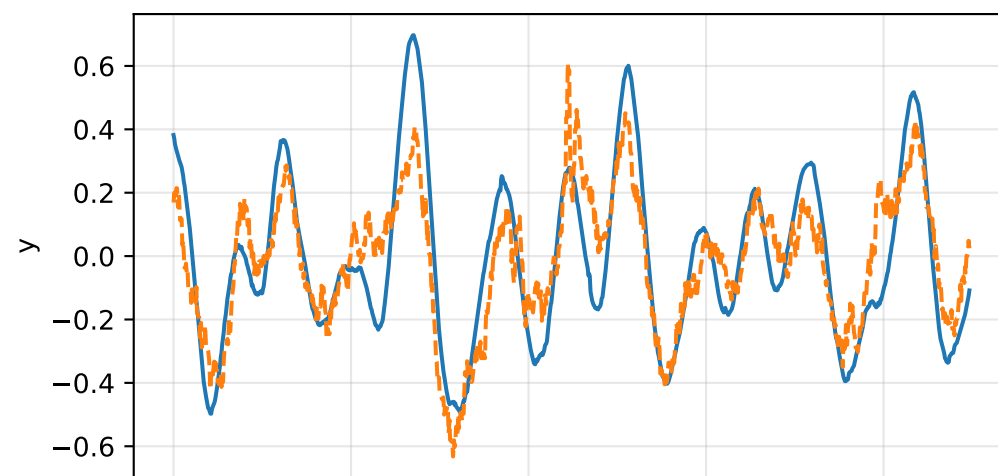
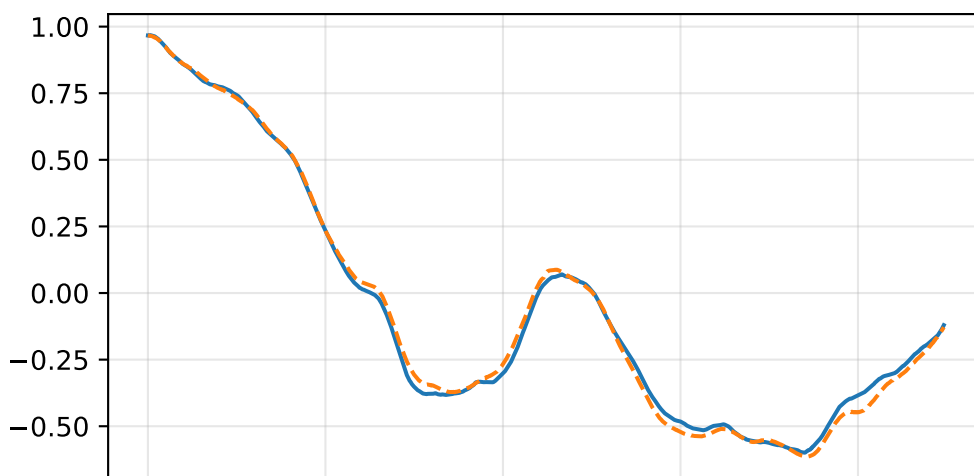
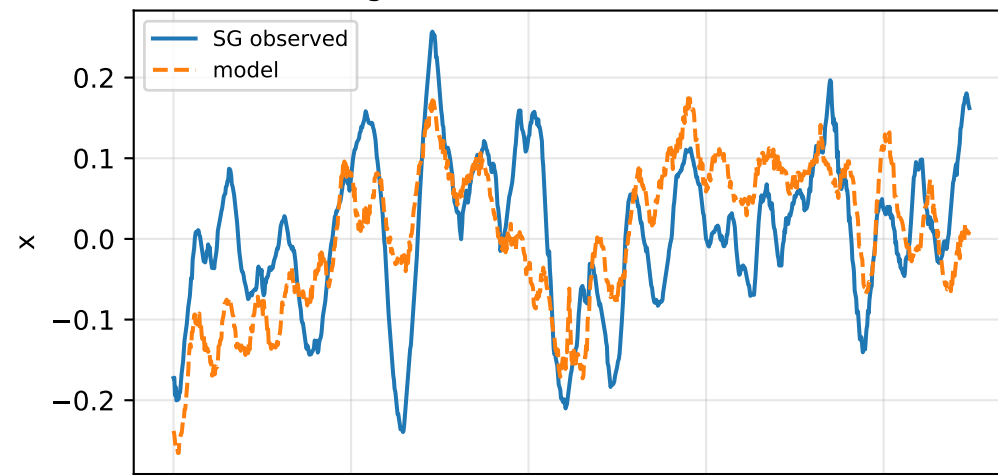


force_over_mass_rotor_2_nominal: acceleration residual objective

specific acceleration [m/s^2]

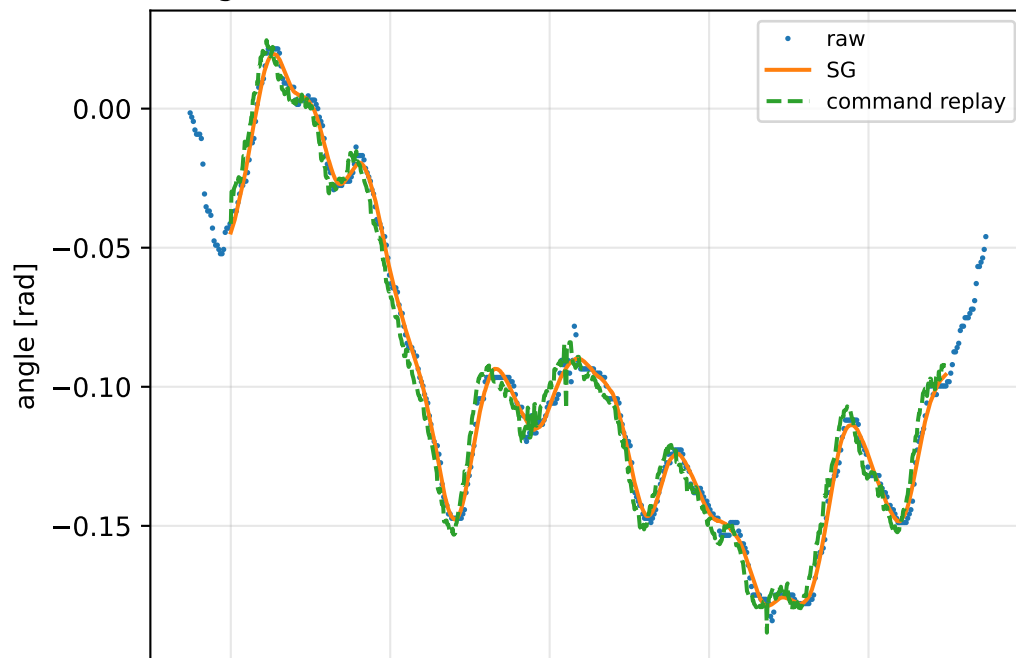


angular acceleration [rad/s^2]

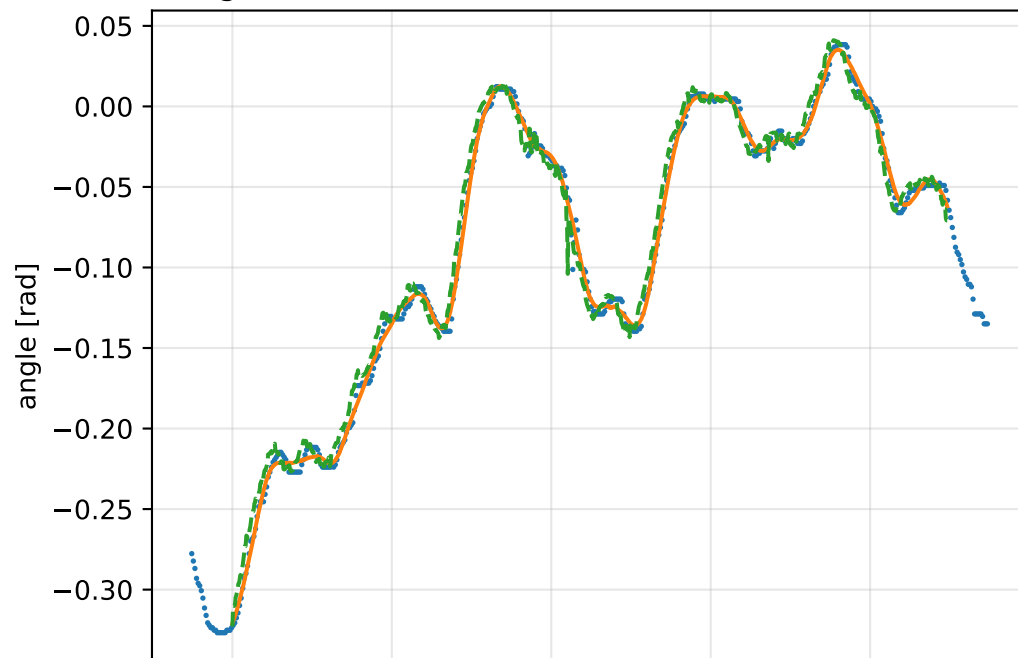


force_over_mass_rotor_2_nominal: gimbal measurement audit (objective=measured_sg)

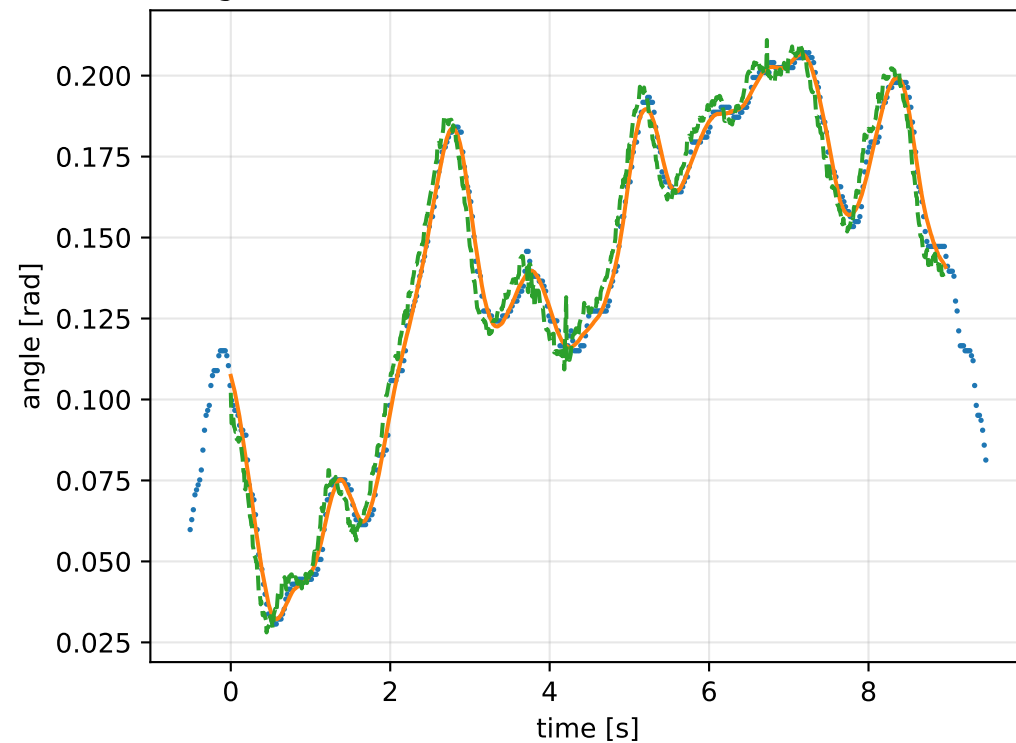
gimbal 1: RMSE=0.006345, max=0.01494 rad



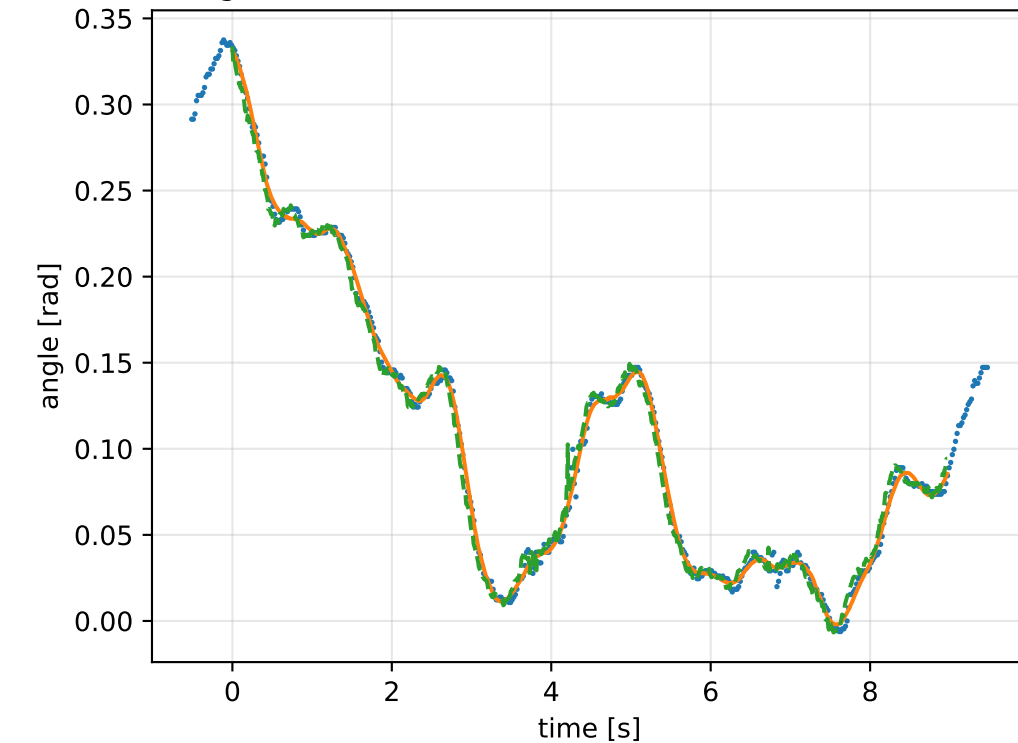
gimbal 2: RMSE=0.008835, max=0.04462 rad



gimbal 3: RMSE=0.007019, max=0.01611 rad

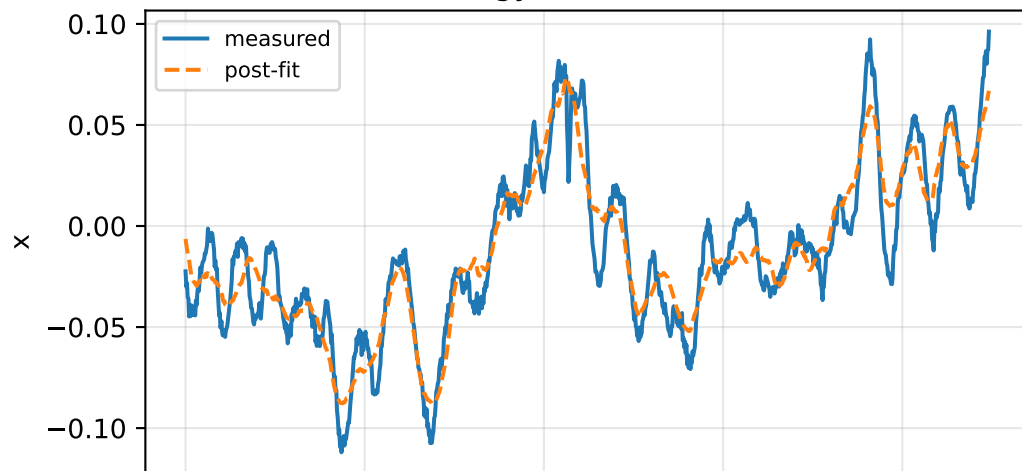


gimbal 4: RMSE=0.006321, max=0.03642 rad

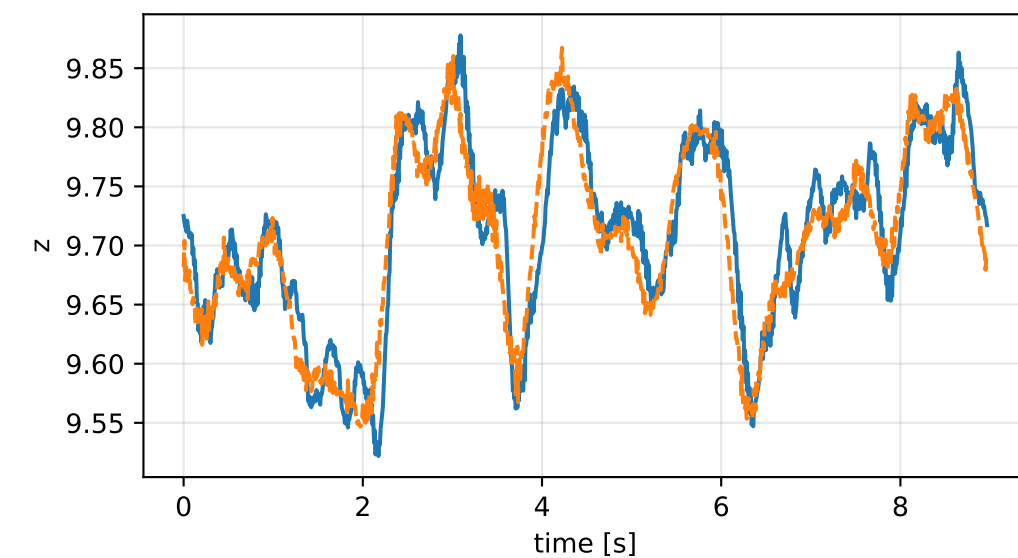
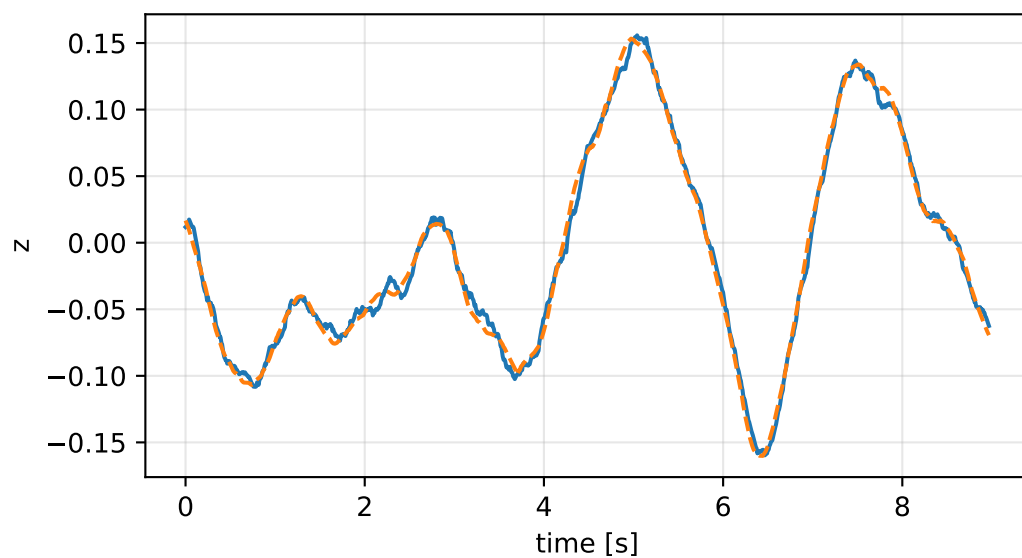
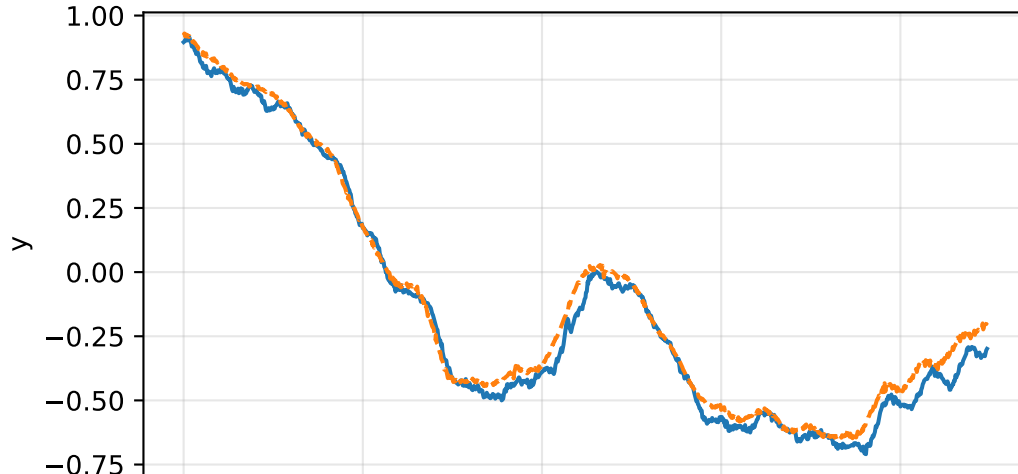
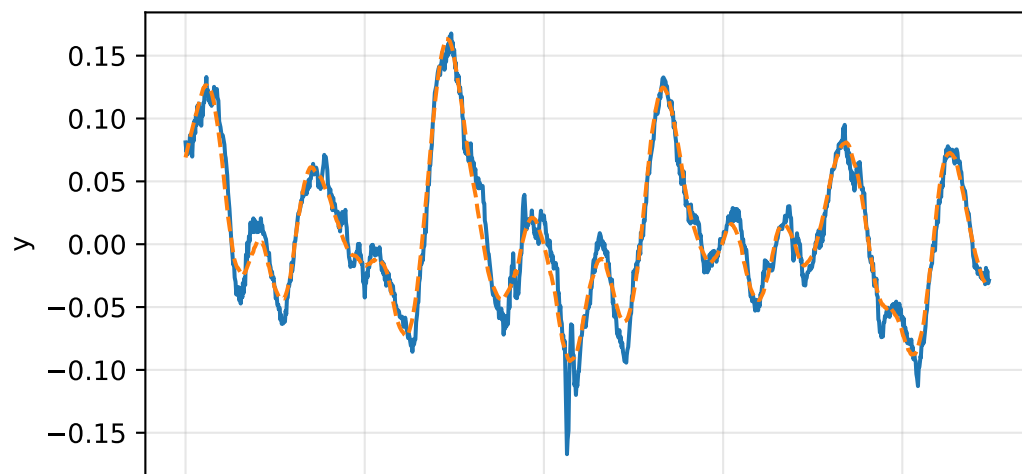
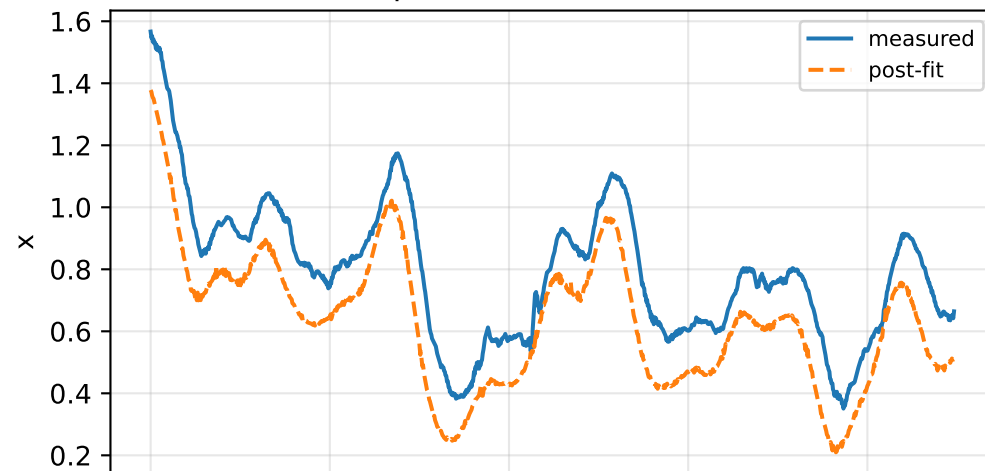


force_over_mass_rotor_2_nominal: IMU comparison (diagnostic only)

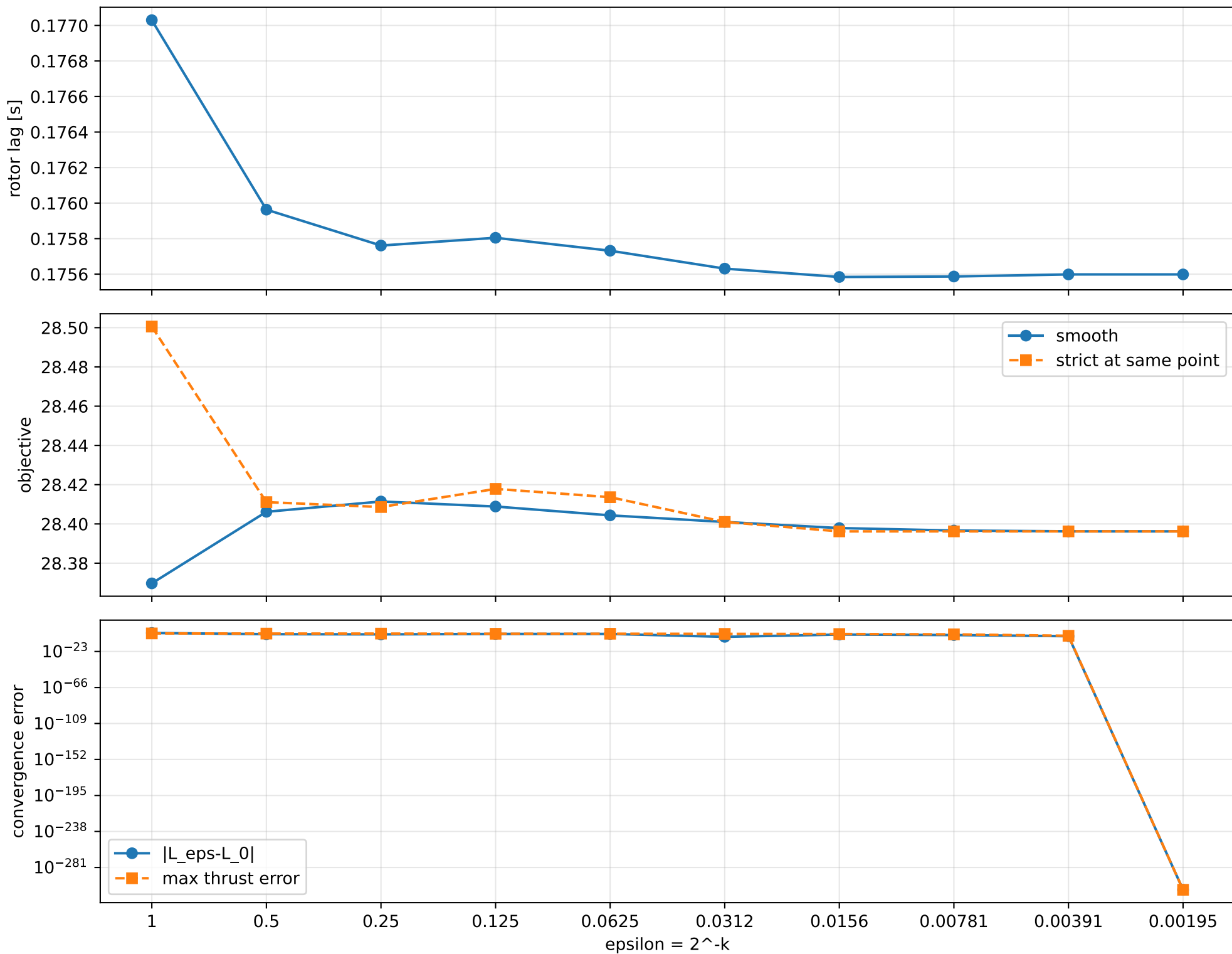
gyro [rad/s]



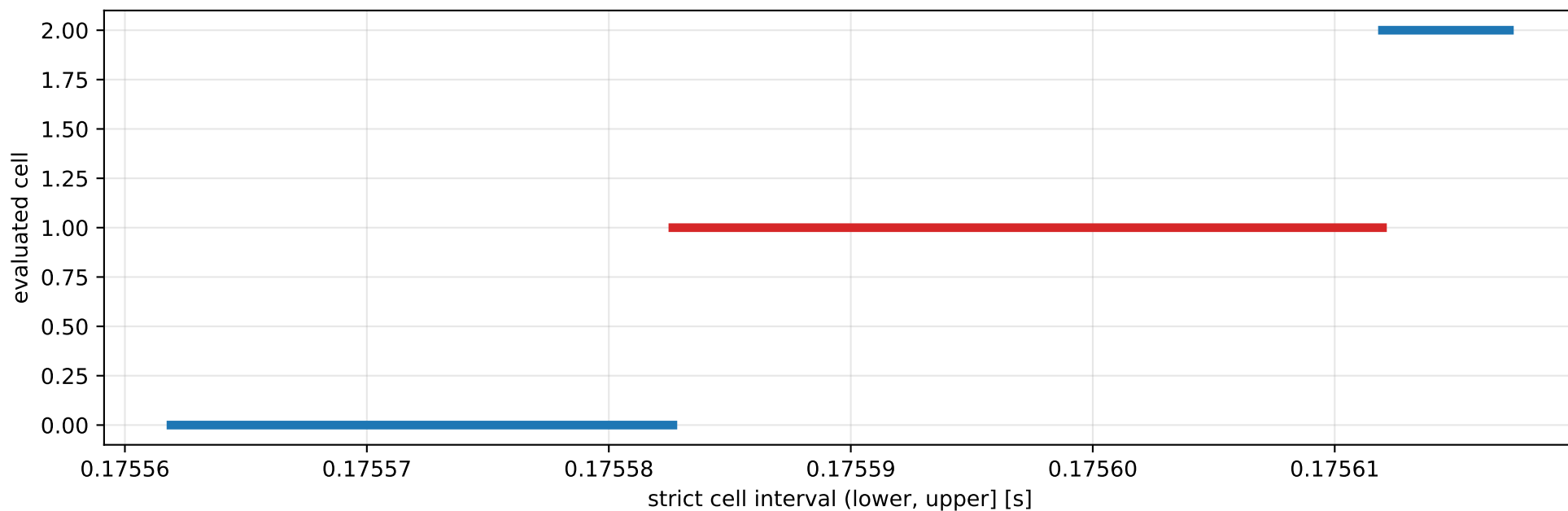
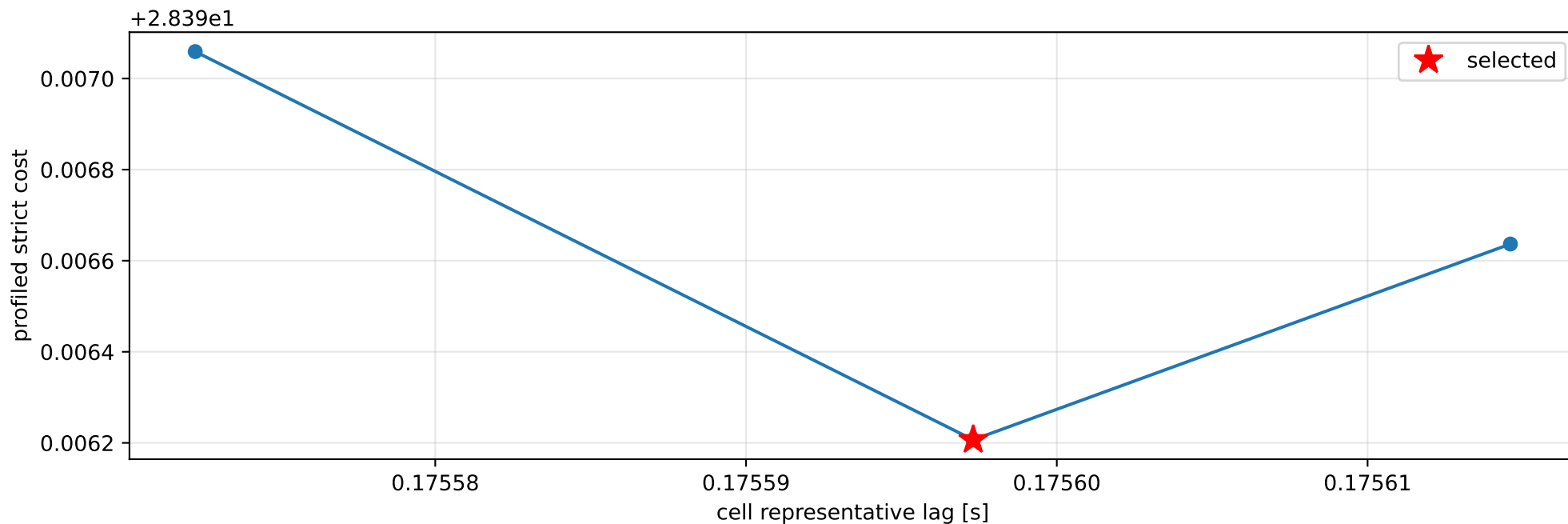
specific force [m/s^2]



force_over_mass_rotor_2_nominal: smooth-to-strict continuation



force_over_mass_rotor_2_nominal: exact strict-ZOH lag cells



selected (0.175582647, 0.175611973] s; final smooth 0.175598355 s; data boundary=False

force_over_mass_rotor_2_nominal: parameter estimate

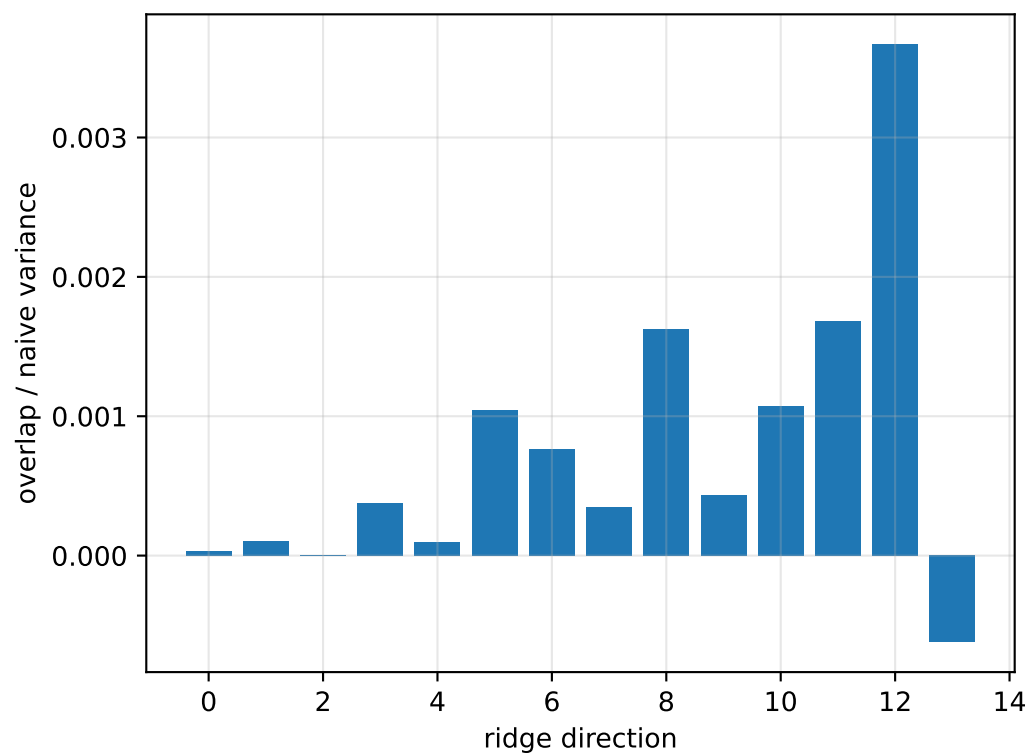
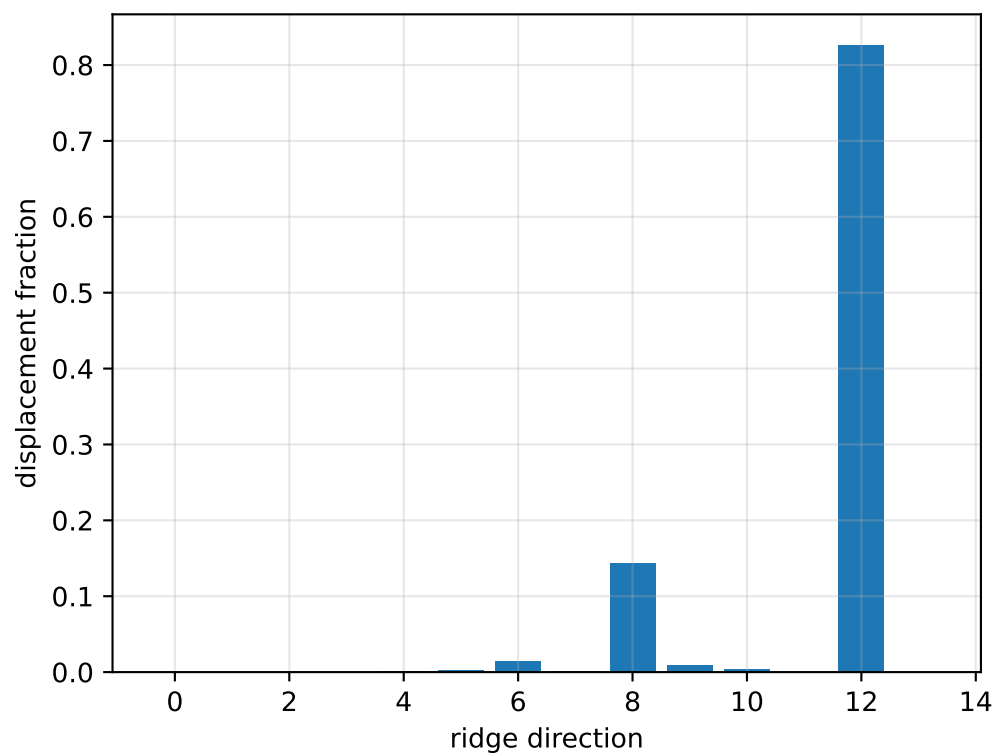
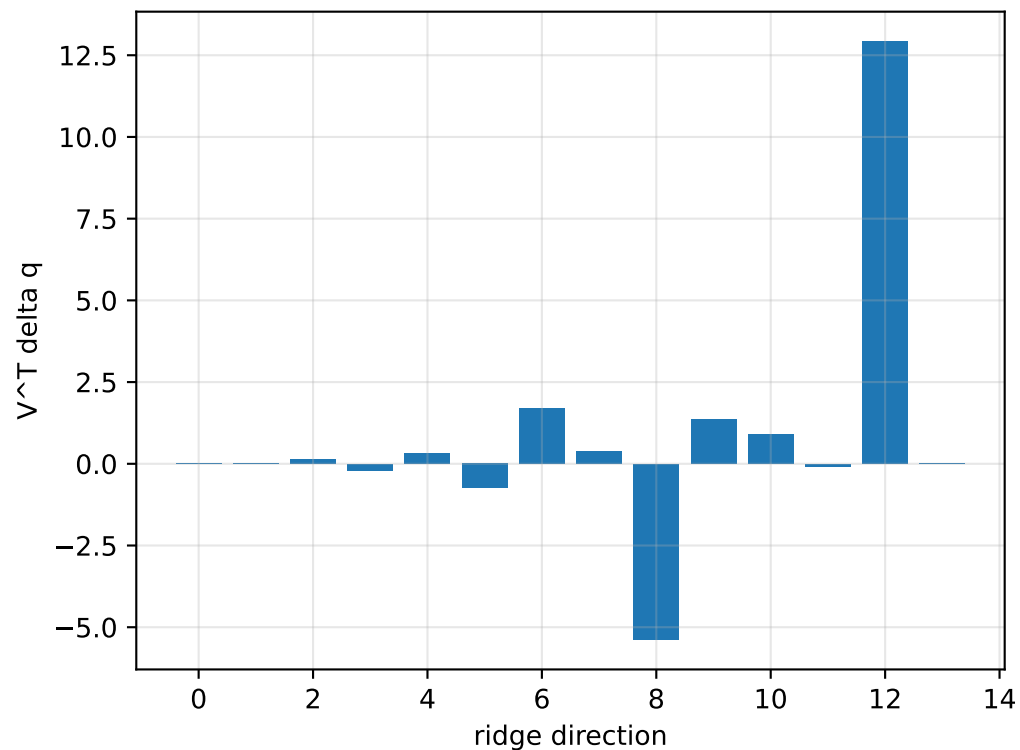
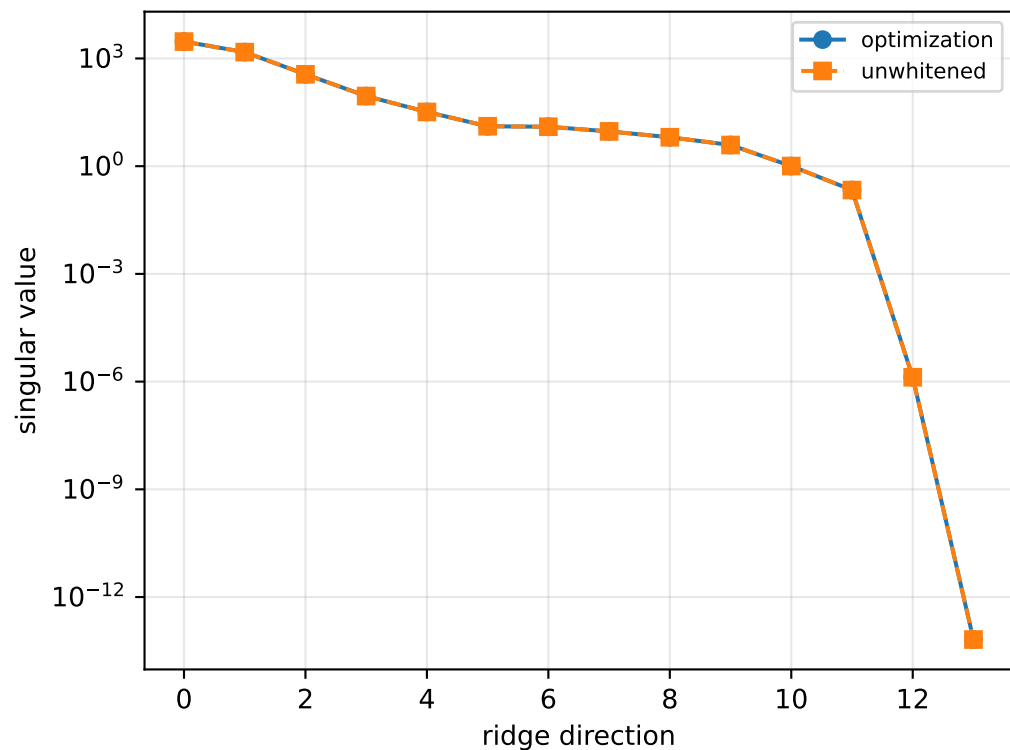
Case: force_over_mass_rotor_2_nominal
status: completed (all stages completed)

mass [kg]: nominal 2.35159759; estimated 5.00293221
inertia [kg m²]:
[[1.15324564 -0.03106627 0.10814213]
 [-0.03106627 0.55111744 -0.01756908]
 [0.10814213 -0.01756908 0.65035349]]
CoG body [m]: [-0.0222089 -0.05317435 -0.00581392]
force effectiveness: [2.23923866 2.12746103 1.1577672 1.4131044]

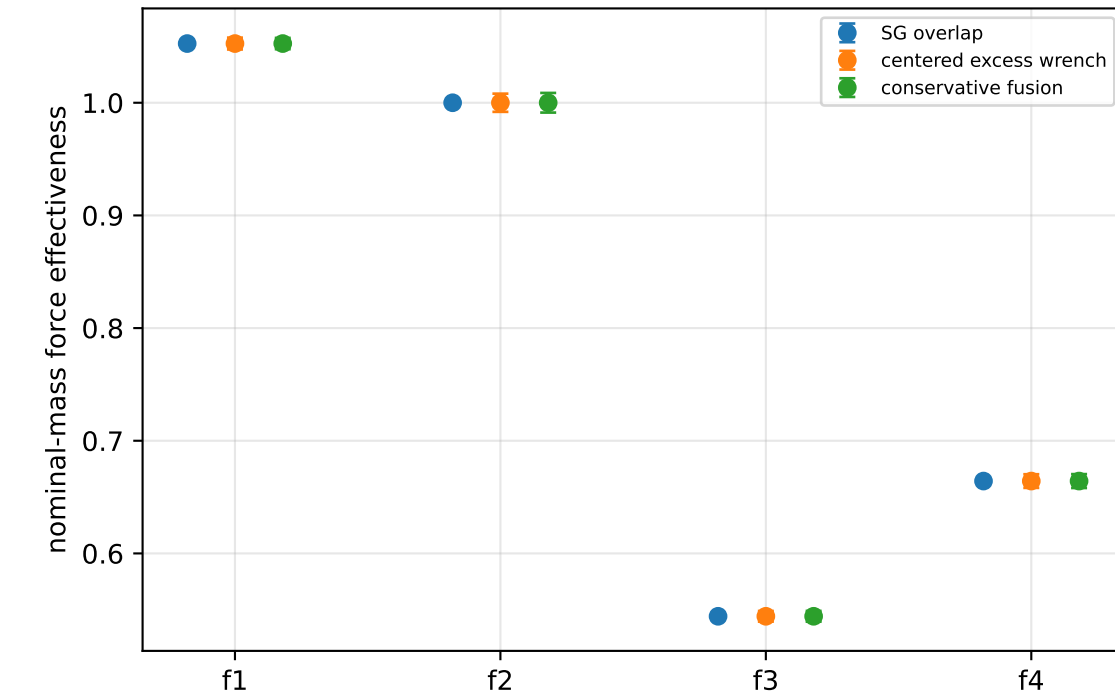
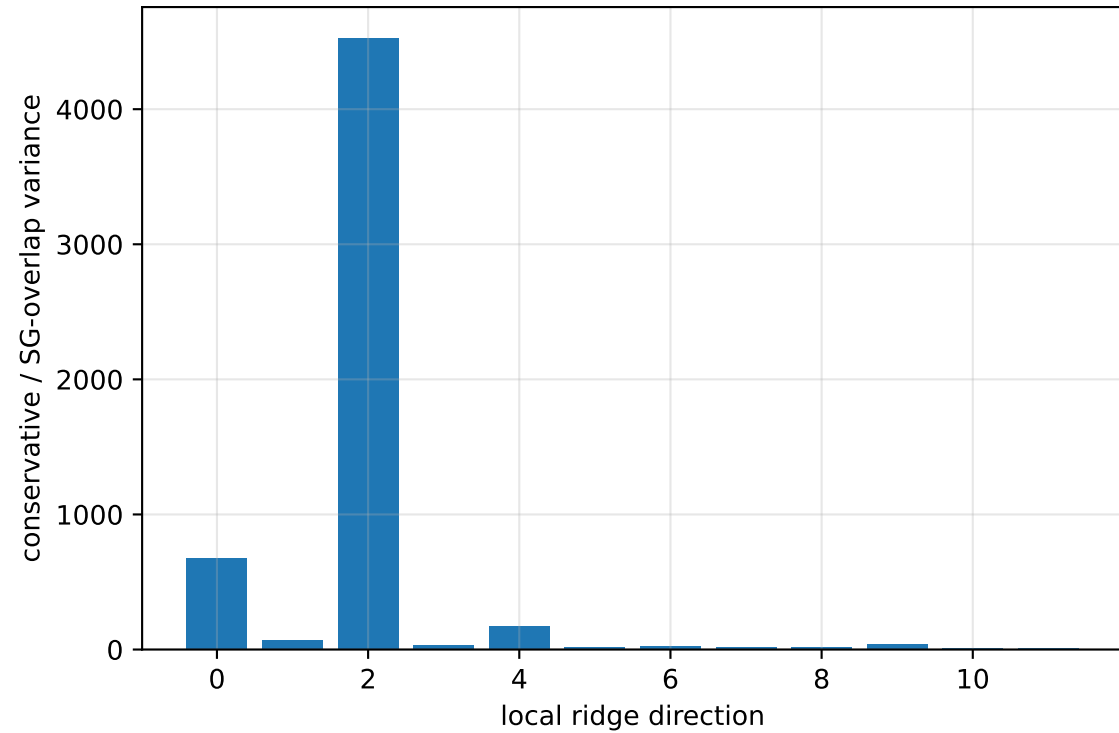
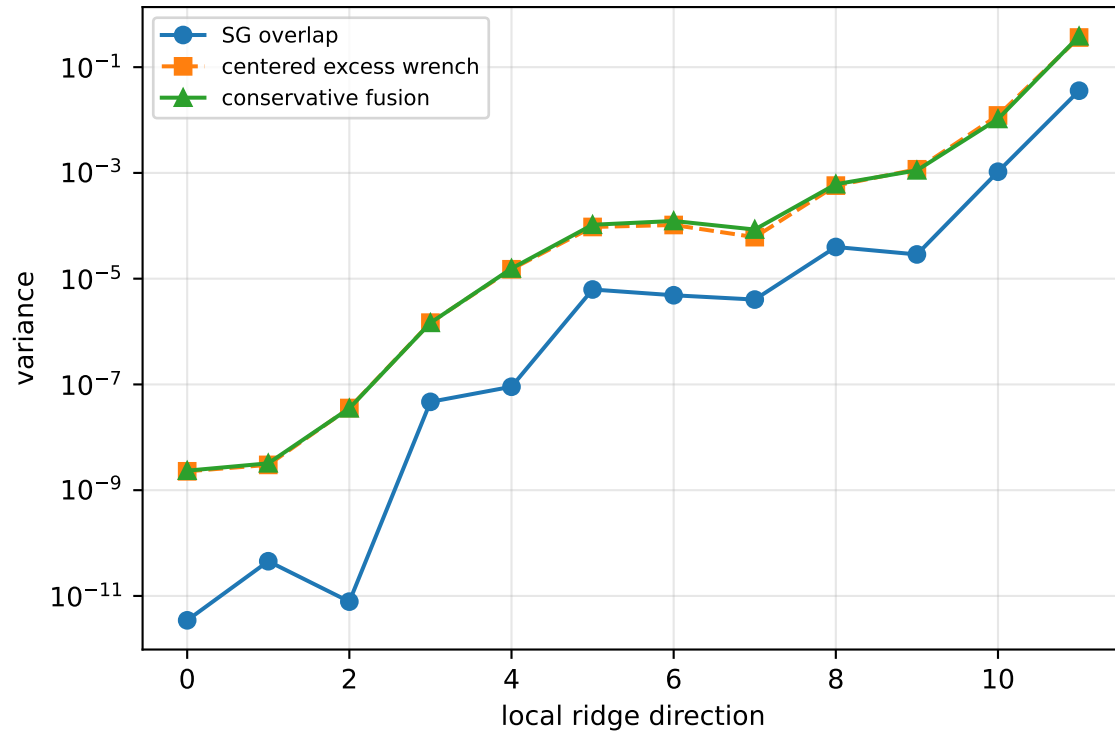
scale-free J/m [m²]:
[[0.23051395 -0.00620961 0.02161575]
 [-0.00620961 0.11015889 -0.00351176]
 [0.02161575 -0.00351176 0.12999446]]
scale-free f/m: [0.44758525 0.42524282 0.23141773 0.28245524]

rotor lag cell: (0.175582647, 0.175611973] s
representative: 0.17559731 s

force_over_mass_rotor_2_nominal: ridge spectrum (rank 13, nullity 1, ||Jv||=7.61e-14)



force_over_mass_rotor_2_nominal: Conservative fusion uncertainty



Top three non-gauge ambiguous directions
 $\text{tr}(\mathbf{M}_{\text{res}}) / \text{tr}(\mathbf{M}_{\text{SG}}) = 390.9$
wrench-to-acceleration recovery max error = $2.92\text{e-}15$
exact scale gauge ridge index = 13 (alignment 1)

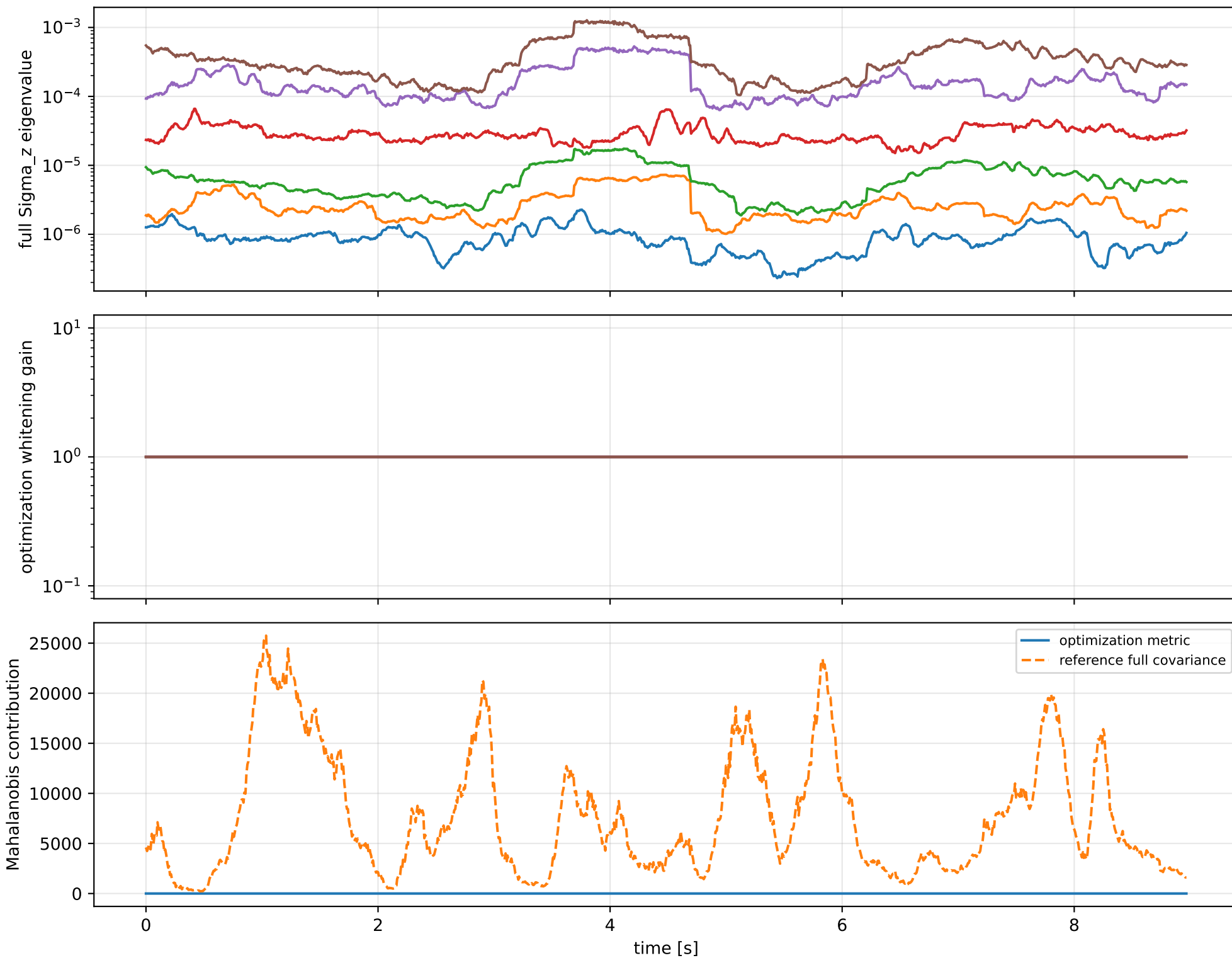
index 11: variance=0.3839, ratio=10.7
 $\log_{\text{mass}}=+0.00216$; second_moment_diag_1=+0.0801
second_moment_diag_2=-0.0746; second_moment_diag_3=-0.0166
second_moment_offdiag_12=+0.989; second_moment_offdiag_13=-0.0897
second_moment_offdiag_23=+0.0293; cog_x=-0.000625
cog_y=+0.000302; cog_z=-0.000386
log_force_eff_1=+0.00408; log_force_eff_2=-0.00242
log_force_eff_3=+0.00198; log_force_eff_4=+0.00534

index 10: variance=0.01048, ratio=9.947
 $\log_{\text{mass}}=-0.0236$; second_moment_diag_1=-0.0171
second_moment_diag_2=-0.0546; second_moment_diag_3=+0.178
second_moment_offdiag_12=+0.0906; second_moment_offdiag_13=+0.973
second_moment_offdiag_23=-0.0526; cog_x=+0.0064
cog_y=-0.000955; cog_z=-0.000631
log_force_eff_1=-0.0411; log_force_eff_2=+0.00379
log_force_eff_3=+0.0175; log_force_eff_4=-0.0632

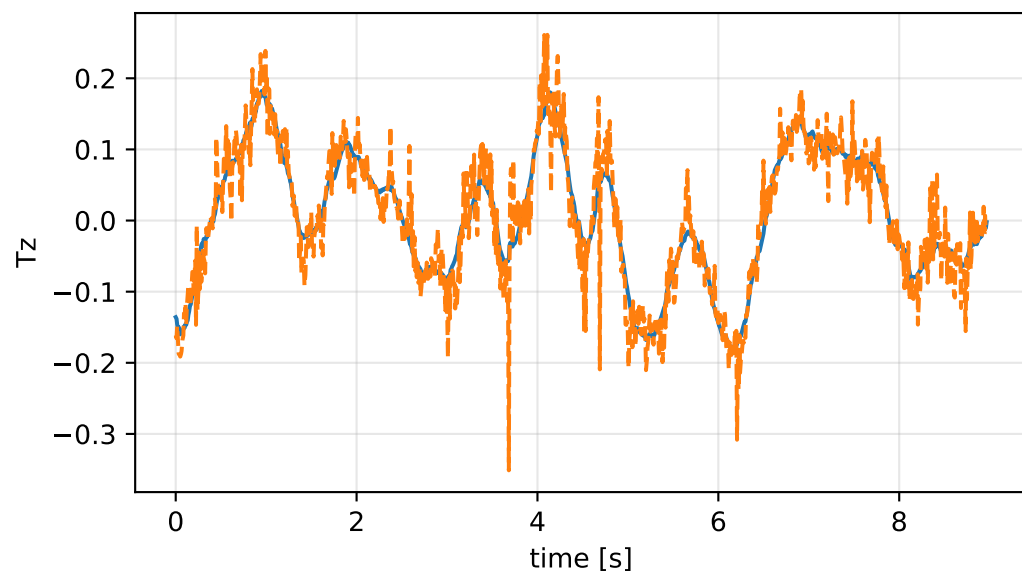
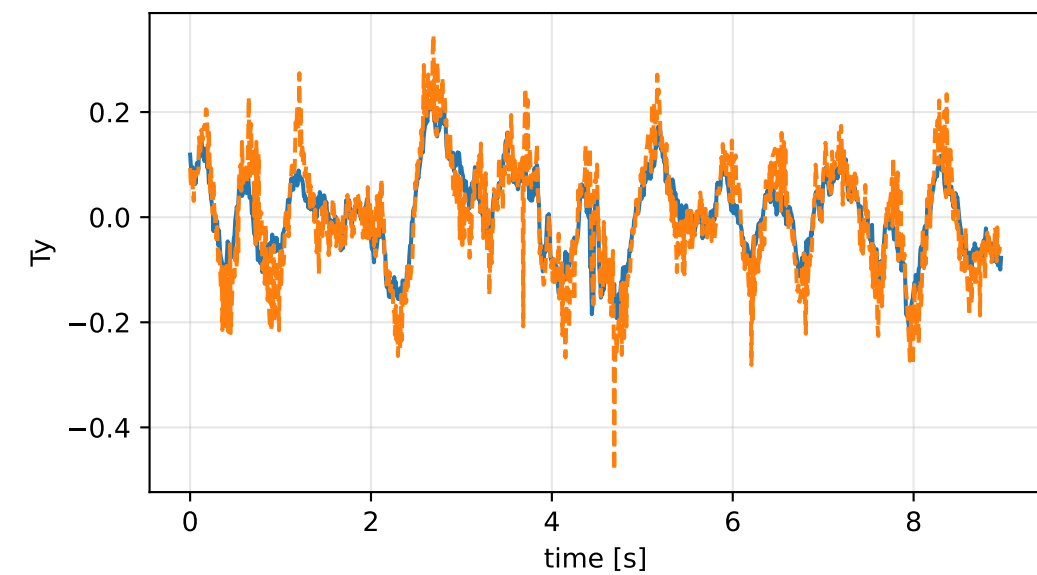
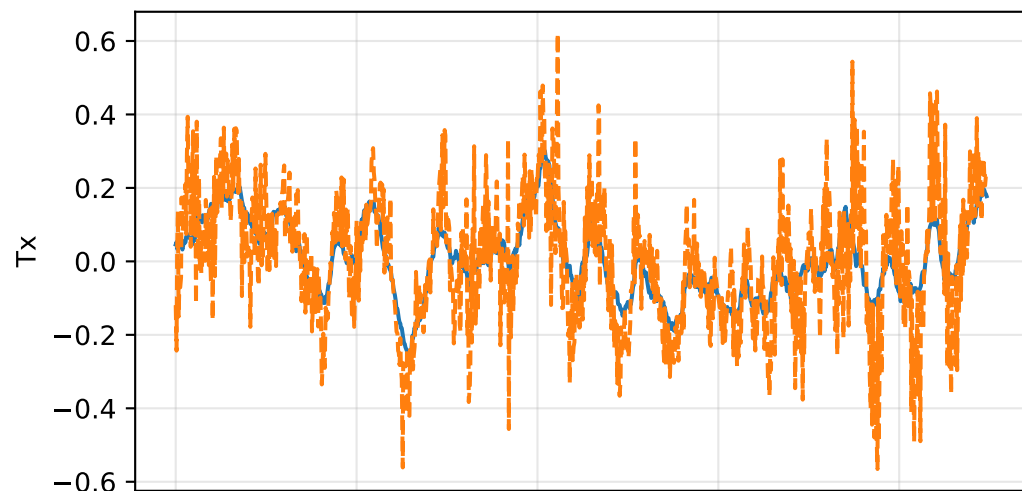
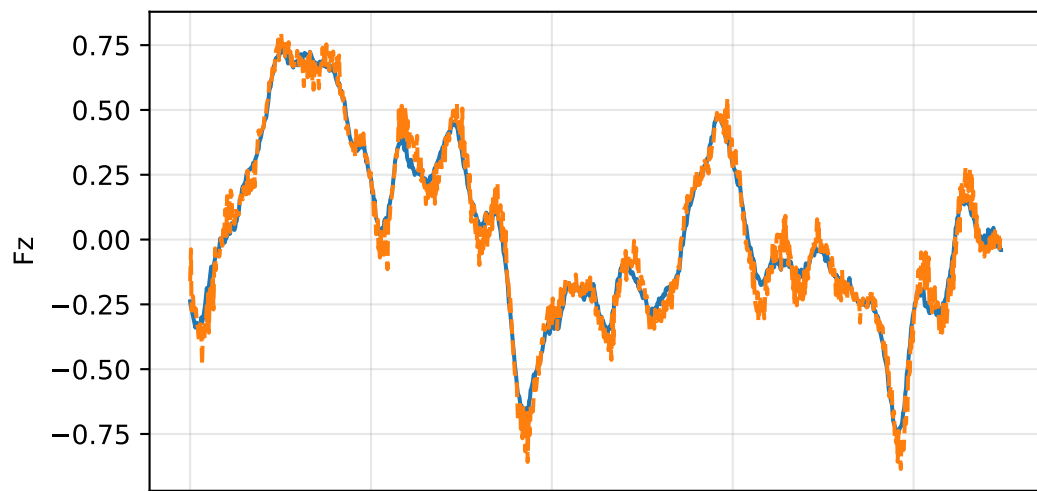
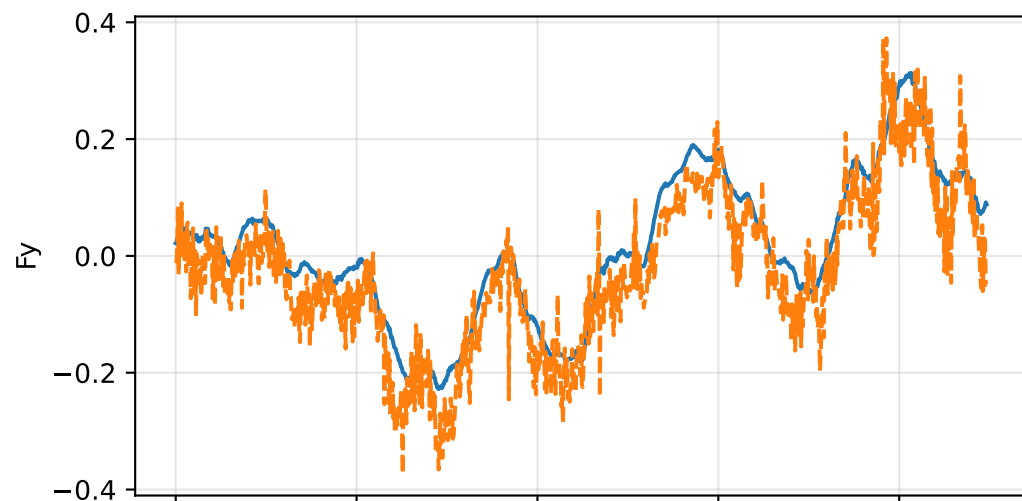
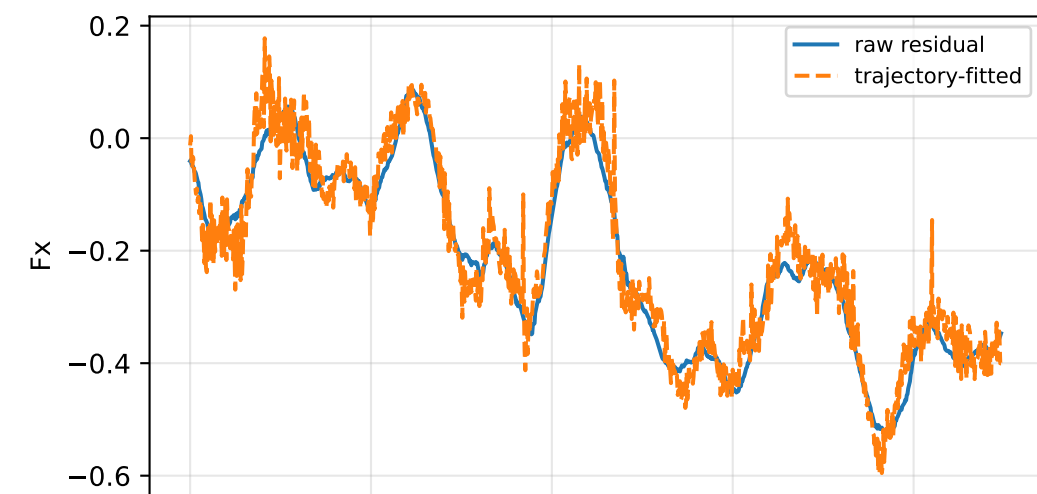
index 9: variance=0.001118, ratio=38.92
 $\log_{\text{mass}}=-0.17$; second_moment_diag_1=+0.00159
second_moment_diag_2=+0.89; second_moment_diag_3=-0.0281
second_moment_offdiag_12=+0.0659; second_moment_offdiag_13=+0.0462
second_moment_offdiag_23=+0.219; cog_x=-0.0064
cog_y=+0.000457; cog_z=-0.0124
log_force_eff_1=-0.16; log_force_eff_2=-0.186
log_force_eff_3=-0.218; log_force_eff_4=-0.129

The conservative fusion covariance deliberately retains the existing SG-overlap uncertainty and adds the uncentered empirical residual score second moment without subtracting the SG contribution. It is intended as a conservative fusion distribution, not as a calibrated generative noise covariance.

force_over_mass_rotor_2_nominal: optimization vs reference covariance



force_over_mass_rotor_2_nominal: wrench / closure (force max $1.46\text{e-}14$, torque max $9.37\text{e-}17$)



force_over_mass_rotor_2_nominal: optional parameter-prior audit

Optional physical parameter prior

The parameter prior is optional external information. The default estimator is prior-free.

name: pseudo_condition_force_over_mass_rotor_2_nominal

role: pseudo_conditioning_ablation

source: /home/leus/catkin_ws/src/ProbTF-demo/ros/examples/grape-param-estim/minimal/config/priors/pseudo_conditioning/scalar/force_over_mass

SHA256: 4ad6b24989a0df886676644a499e15946486f27db58e2dfd259bd888c891980e

data objective: 28.3962067786

prior objective: 6.13201011788e-09

total objective: 28.3962067847

This configuration uses an intentionally tight artificial standard deviation for an ablation-style pseudo-conditioning experiment; it is not a calibrated physical uncertainty.

factor: force_over_mass_rotor_2_nominal (force_effectiveness_over_mass)

components: rotor_2

target: [0.42524282]

std: [1.e-05]

final: [0.42524282]

error: [1.10743037e-09]

standardized residual: [0.00011074]

factor objective: 6.13201011788e-09